Muhammad Rishal P

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PROFESSIONAL SUMMARY

Computer Science & Engineering undergraduate specializing in systems programming, local AI & RAG architectures, terminal tooling, and embedded IoT. Proficient in Python, TypeScript, C/C++, and Java, with hands-on experience in full-stack architectures, low-level system implementations, and government IT operations.

EDUCATION

Government Engineering College, Kannur — **2024 – Present**
B.Tech in Computer Science and Engineering — *Kannur, Kerala*

- Relevant Coursework: Data Structures, Object-Oriented Programming, Database Systems, Operating Systems.

E.K. Nayanar Memorial Model Polytechnic College — **2020 – 2023**
Diploma in Electronics and Communication Engineering (CGPA: 8.77) — *Kalyassery, Kerala*

- Strong foundation in digital electronics, networking protocols, and low-level embedded software.

EXPERIENCE

Infosys Springboard — **2025 – 2026**
Software Development Intern (Virtual) — *Remote*

- Developing a Resume Analysis & Job Recommendation engine using Python, Streamlit, and LangChain parsing.
- Implemented secure user authentication with bcrypt and persistent SQLite3 storage.

KELTRON (Kerala State Electronics Development Corporation) — **2023**
Network Supervisor & Digital Signature Project Coordinator — *Kannur, Kerala*

- Managed IT and network operations at the Department of Treasuries, Government of Kerala; coordinated DSC implementation.

SFO Technologies — **2023**
Production Technician – Wire Harnessing — *Kochi, Kerala*

- Assembled and inspected high-reliability wire harnesses following industrial quality standards.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, C / C++, Java, SQL, NASM Assembly
- **Systems & AI:** Local LLMs (Ollama GPU), SQLite FTS5 RAG, Piper TTS (ONNX), Web Audio API, Ratatui TUI
- **Backend & Cloud:** FastAPI, Node.js, Express.js, REST APIs, Docker, SQLite (WAL Mode), PostgreSQL
- **Hardware & IoT:** ESP32, Tasmota, MQTT (Mosquitto), Linux (Hyprland / Wayland), Embedded C, Flutter

KEY PROJECTS

J.A.R.V.I.S. — Voice AI Companion — **github.com/rishalsha/Jarvis**
`Python • Ollama GPU (RTX 3060) • Piper TTS (75ms) • SQLite FTS5 RAG • Web Audio HUD • MQTT`

- Private voice AI with local GPU inference, 75ms neural TTS, sub-1ms SQLite FTS5 memory recall, and barge-in detection.

papr — Terminal Typesetting Engine — **Crates.io: papr-cli**
`Rust • CLI • Tectonic • XeTeX • KaTeX`

- Published on Crates.io: compiles Markdown, multi-file LaTeX, and plain text into publication-grade PDFs with auto package resolution.

qmit — AI Conventional Commit CLI — **Crates.io: qmit**
`Rust • LLM APIs • Git CLI`

- Published on Crates.io: inspects staged Git diffs and generates standardized Conventional Commit messages (v1.0.0).

IPadDock — Mission Control Battle Station — **github.com/rishalsha/IPadDock**
`Python • Mosquitto MQTT • Linux Hyprland/Wayland IPC • Pure ES5 SVG`

- Repurposed iPad Gen 4 (iOS 10.3) into a permanent battle-station HUD, multi-machine stream deck, and MQTT telemetry center.

JsonBin — Self-Hosted JSON Engine — **github.com/rishalsha/JsonBin**
`FastAPI • Python 3.12 • SQLite WAL • Docker`

- Self-hosted JSON storage backend compatible with JSONBin.io v3 API with SQLite WAL persistence and Docker packaging.

GPT-CLI — AI Terminal Assistant — **github.com/rishalsha/GPT-CLI**
`Go • Markdown Streaming • Unix Piping • LLM APIs`

- High-speed terminal AI assistant featuring real-time Markdown streaming, multi-turn chat sessions, and Unix pipe support.

GitHubCLI (ghcli) — GitHub Management TUI — **github.com/rishalsha/GitHubCLI**
`Rust • Ratatui • Crossterm • GitHub REST API`

- Keyboard-driven Ratatui terminal UI for managing GitHub repository lifecycles (create, clone, search, delete) from the shell.

AI Resume Analysis & Portfolio Generator — **github.com/rishalsha/Project-Mini**
`Python • Streamlit • LangChain • SQLite3 • bcrypt`

- Parses candidate resumes, extracts skill graphs, matches job opportunities, and generates automated responsive portfolios.

EasyBin — Web Paste & Storage Engine — **github.com/rishalsha/EasyBin**
`JavaScript • Node.js • REST API`

- Lightweight pastebin and payload storage service for sharing code snippets and JSON documents with instant lookup.